Predict the crash.
Price the fix.

A remote-operations digital twin for haemodialysis. It predicts intradialytic hypotension 15–60 minutes ahead, then shows what each fix costs — in blood pressure, and in fluid left behind.

303,122
dialysis sessions

2
countries

1,828
patients

[Team name] · [College] · [Member 1] · [Member 2] · [Member 3]

THE PROBLEM

A dialysis session is a four-hour tightrope

Remove too little fluid and the patient stays overloaded. Remove it too fast and the pressure falls. A systolic below **90 mmHg** during the session is intradialytic hypotension — **IDH**.

2× vascular-access thrombosis in the highest IDH quartile
Chang, JASN 2011

9.4× intradialytic arrhythmia with IDH
Pun, CJASN 2021

1.57 hazard ratio for mortality when nadir systolic falls below 90
Flythe, NDT 2018 · association, not proof of causation

The event takes minutes.
The measurement takes half an hour.

One session · one cuff every ~30 minutes

Schematic. Nine readings across four hours, and an event between two of them.



The monitor alarms when the pressure is already low. A forecast is a different object: it has to arrive before the gap closes.

WHY INDIA

1,700+ district centres. No nephrologist on site.

“This is the terminal ward of India’s T2D epidemic.”

34.5% of our dialysis cohort is diabetic (370 of 1,072 patients). India has about **101.3 million** people with diabetes and 136 million with prediabetes (ICMR-INDIAB 2023). Kidney failure is where that epidemic ends — on a machine, three times a week.

One nephrologist cannot be in 1,700 buildings.
Remote supervision is not a feature. It is the only arithmetic that works.

1,700+

dialysis centres under the Pradhan Mantri National Dialysis Programme, many in district hospitals without on-site nephrology → remote supervision is the only scalable model

156

sessions per patient per year — three a week, for a median dialysis vintage of 9.6 years → no home sensor produces this density

~34M

dialysis sessions a year in India → an estimate, labelled as one

The predictor is published. We are not claiming it.

PUBLISHED · DEPLOYED · NOT OURS

Zhang et al. · *Nephrology Dialysis Transplantation* · 2023

0.89
AUROC, 15–75 min ahead

15–75
minute horizon

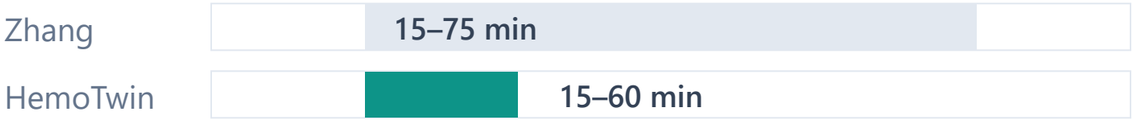
6
US clinics, live

42,656
sessions, 693 patients

Same family of model as ours: boosted trees on machine telemetry. The difference is what happens after the score.

It works, it is in clinics, and a judge has seen it before. So we built the layer it does not have.

We do not claim the predictor.



01 **The UF-policy twin.** Hold, taper, stop — each simulated forward, each with its fluid cost attached.

02 **A volume state with ground truth.** Modelled ultrafiltration, checked against measured weight loss.

03 **Two countries of data.** 303,122 sessions across Taiwan and Spain — and a console built for one nephrologist covering many chairs.

Our horizon is narrower, and a narrower horizon is an easier task. We do not compare the two AUROCs — we compare on what the output lets a nurse do.

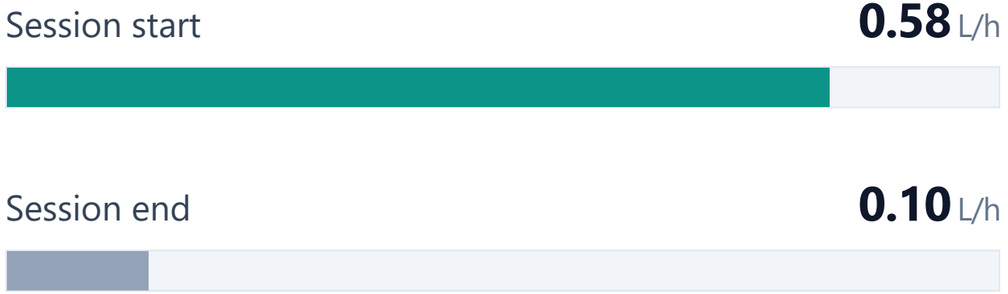
THE INSIGHT

Everyone already tapers. They just guess when.

87.2%

of **32,565 real sessions** cut ultrafiltration by more than 20% mid-session. **81.7%** end below 80% of their peak rate.

MEDIAN UF RATE · MEASURED



A 5.8-fold fall in the median rate. The staff are not waiting for the alarm — they are already reacting.

The question was never whether to taper.
It is when — and what it costs.

THE REFRAME

Predict the crash. Price the fix.

COST ONE

Pressure too low, now

Access thrombosis, arrhythmia, a session that ends in saline and a stretcher.

COST TWO

Fluid left behind, later

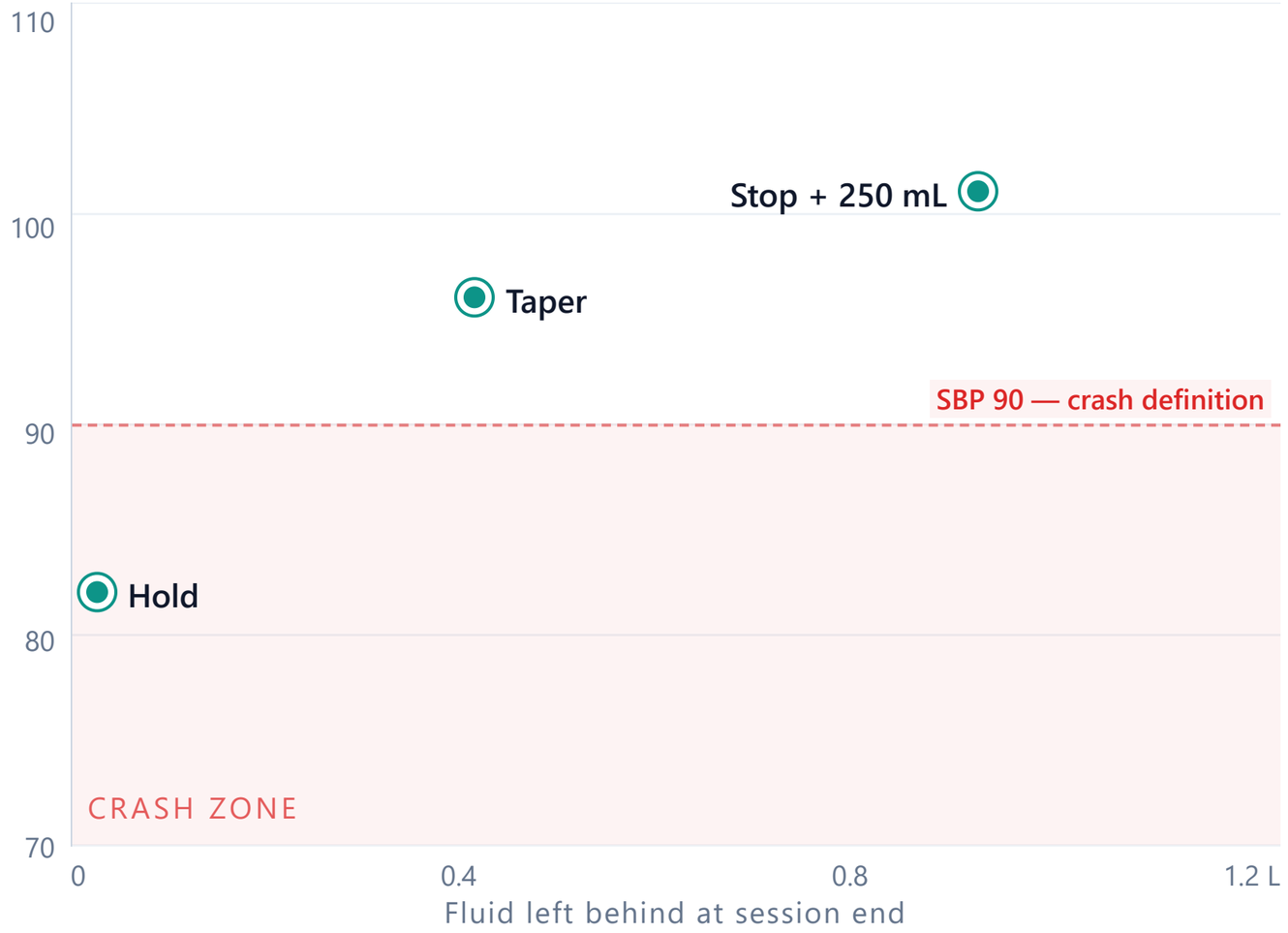
Overload, a harder next session, and the same trade-off again on Thursday.

WHAT THE TWIN ADDS

Both axes, one picture

Each policy becomes a point: nadir on the vertical, litres left on the horizontal.

> **A risk score has one number. A decision has a price.**



WHAT WE BUILT

Three layers, one decision

MEDICAL IOT 4,190,926 timestamped rows · 2,342,586 BP events · UF rate, blood flow, venous pressure

EHR age · sex · diabetes · dialysis vintage (median 9.6 years) — mapped to FHIR / ABDM at deployment

Layer 01

Event model

40 features, patient-disjoint split. Answers one question: will systolic fall below 90 in the next 15–60 minutes?

AUROC 0.916 · catches 75.7% of crash sessions

→

Layer 02

Physiological twin

Holds a volume state for this patient, with two named parameters. Predicts the next BP, then corrects when the cuff fires.

$V(t) \cdot k_{refill} \cdot \beta_{bp}$

→

Layer 03

Counterfactual engine

Projects pressure over the next hour under hold, taper and stop, and does the fluid accounting to session end. Returns a nadir with an interval, and the fluid each policy leaves behind.

3 policies · empirical band · litres left

ACTION SHEET Taper to X mL/h · recheck in 15 minutes · escalate to the remote nephrologist if nadir < 90 mmHg. Nothing fires automatically.

VALIDATION · EVENT LEVEL

0.916, and the baselines that keep us honest

AUROC

0.916

95% CI 0.897–0.932
patient-clustered bootstrap

AUPRC

8.4 × lift

AUPRC 0.236 against 2.80% test prevalence

CRASH SESSIONS CAUGHT

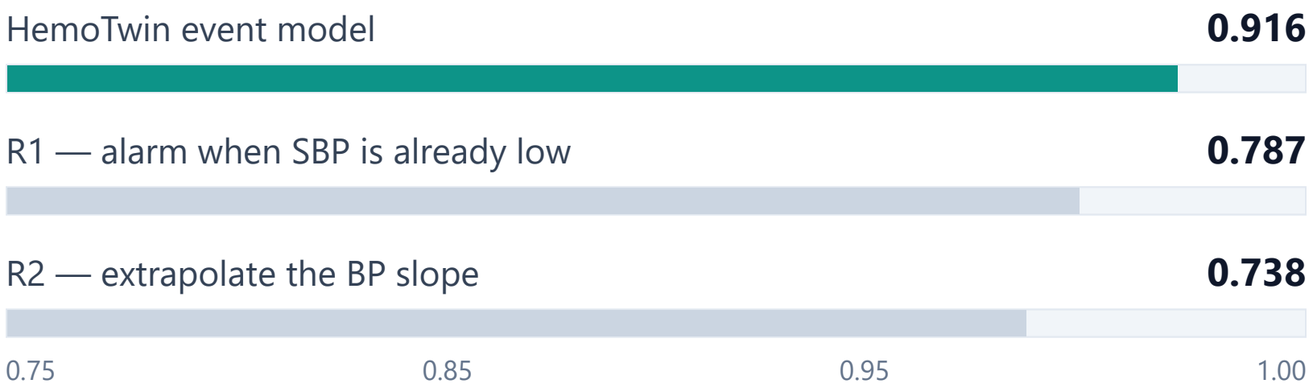
75.7 %

1,783 of 2,354, at least 15 min before onset

FALSE-ALARM SESSIONS

20.6 %

sessions without IDH that raise an alarm



WHAT THE HEADLINE DOES NOT SAY

86.0% of crash sessions with a usable warning window (1,783 / 2,073). **281** sessions had no anchor in the window — a cuffing-cadence limit, reported not hidden.

1,268,990 prediction anchors, 32,221 positives (2.54%), 1,070 patients, split by patient. And our 15–60 min AUROC is **not** comparable to a 15–75 min result.

Our AUPRC of **0.236** is higher than the best published external AUPRC (Kim 2022, 0.106), on a different cohort and a narrower horizon. We do not treat that as a like-for-like win. We lead with **detection and alarm burden**, not the area under the curve.

Two equations a nephrologist can argue with

STATE · VOLUME

$$\frac{dV}{dt} = -UF(t) + k_{refill} \cdot (V_{target} - V(t))$$

Ultrafiltration pulls volume out. Tissue refills plasma at a rate set by **k_{refill}**. The deeper the deficit, the harder the refill pushes back. That is the whole volume model.

k_{refill} — per hour · starts from a population prior, updated from the patient's own sessions

OUTPUT · PRESSURE

$$SBP(t) = SBP_0 + \beta_{bp} \cdot (V(t) - V_0) - \gamma \cdot elapsed(t)$$

Systolic pressure rides on the volume still in the patient, scaled by **β_{bp}**, minus a slow drift with time on the machine.

β_{bp} — mmHg per litre · the number that differs most between patients

PREDICT
Forward to the next cuff

→

MEASURE
The next real BP arrives

→

CORRECT
State and both parameters update

Two personal parameters, not twenty. With a cuff every ~30 minutes, anything richer is not identifiable from one session — it is curve-fitting with a nicer diagram. We chose the model a nephrologist can audit, then checked its volume state against measured weight loss.

THE WHAT-IF

Chair 1, minute 110. Three choices, one price list.

A · HOLD — CRASH

Hold current UF

70 mmHg

90% interval 33–92 mmHg

Fluid left at end **0.0 L**

Target met, and systolic reaches 70. The real onset was minute 140. Saline and a UF stop follow.

B · TAPER — THE CHEAP FIX

Taper UF by 25%

77 mmHg

90% interval 40–99 mmHg

Fluid left at end **0.45 L**

Not enough. The point estimate still falls below 90, and 0.45 L stays behind.

C · STOP — THE RESCUE

Stop UF + 250 mL

103 mmHg

90% interval 67–126 mmHg

Fluid left at end **1.79 L**

Clears the threshold. The price is 1.79 L of prescribed fluid not removed.



The cheap fix is not enough. The fix that works costs 1.8 litres. That is the decision.

NEVER ONE BLENDED AUROC

Two cohorts, two tasks, no blended number

HEMOBP · Taiwan	event level
Task	**15–60 min ahead**
Patients	**1,070**
Sessions	**205,482**
BP events	**2,342,586**
Prediction anchors	**1,268,990**
Positives	**32,221 · 2.54%**
Split	**patient-disjoint**
AUROC	**0.916 [0.897–0.932]**

CC BY 4.0 · Lin et al., *Scientific Data* 2019

Mendeley · Spain	session level
Task	**session-level hypotension?**
Patients	**758**
Sessions	**97,640**
Prevalence	**25.5% · 24,912**
Label	**symptomatic · SI / NO**
Granularity	**one pre + one post BP**
Time series	**none — cannot test 15–60**
Transport	**0.474 — failed**

CC BY 4.0 · external frame only; transport failed, so no external claim

VOLUME-STATE CHECK

Modelled ultrafiltration versus measured weight loss (start minus end): median difference **+0.07 L**, MAE **0.40 L**, and **75%** of sessions within **0.5 L**. Measured UF median **2.2 L** per session across 16,591 sessions. The state variable is checked, not assumed.

TRUST

Why a nurse should believe the number

01 · CALIBRATION

Probabilities, not scores

A reliability diagram, a Brier score, and the cohort base rate printed beside every risk. A 50% that means 50 out of 100.

02 · ALARM BUDGET

The nursing team picks the trade

Threshold fixed on validation at 0.0410. On test: sensitivity 0.862, **20.6%** false-alarm session rate. The operating point is a staffing decision, and it is shown, not buried.

03 · DRIVERS

SHAP, translated

“UF rate above her usual.” “One hundred and ten minutes into the session.” “Refill index below her own baseline.” Reasons a nurse can check.

04 · AUDIT

Every override is data

Recommendation, override, recheck and outcome are timestamped and exportable. Overrides are reviewed weekly — they mark where the model and the ward disagree.

05 · SUBGROUPS

Pre-specified, not yet run

Vintage, diabetes, sex, age band. The analysis is specified in the model card and will be published with intervals — including where it is worse. We do not claim fairness we have not measured.

LIMITATIONS

What this cannot do yet

One cohort; the external check failed

Taiwanese data only, internally validated. A session model scored **0.720** internally and **0.474** in Spain, because the labels mean different things. No Indian cohort. Deployment means retraining, never transplanting probabilities.

Failure reported in docs/EXTERNAL_VALIDATION.md

A cuff every ~30 minutes

281 crash sessions had no anchor inside a warnable window. The model cannot see what the machine does not measure.

Reported as 75.7% of all, 86.0% of warnable

“SBP under 90” is a definition

Not a symptom. We predict a number crossing a threshold, not dizziness, cramps or a patient who feels awful.

Symptom labels are not in these cohorts

The counterfactual is observational

Within-policy-space simulation: everyone tapers, so we are choosing timing, not inventing an intervention. Validated on held-out patients, with intervals.

A decision aid, not a prescription

Not a medical device

No regulatory clearance. No diagnosis, no prescription, no automatic action. It is a research prototype in front of clinicians.

CDSCO pathway not started

Alarm burden is real

20.6% of uncomplicated sessions raise an alarm. That is a ward cost, and the reason the threshold is a nursing decision.

Tunable; shown, not hidden

WHAT NEXT

Predict the crash.
Price the fix.

01 · Now

Shadow mode

Run on a partner machine, silent. Log predictions, overrides and outcomes. No clinical action.

02 · Next

Prospective comparison

Pre-registered, against standard care. Measure IDH rate, alarm burden, fluid removed.

03 · Then

Retrain in India

Local data, local thresholds, FHIR / ABDM. The model travels as a method, not as weights.

[Team name] · [College] · [Member 1] · [Member 2] · [Member 3] · [contact]